

Pit Planning and Design

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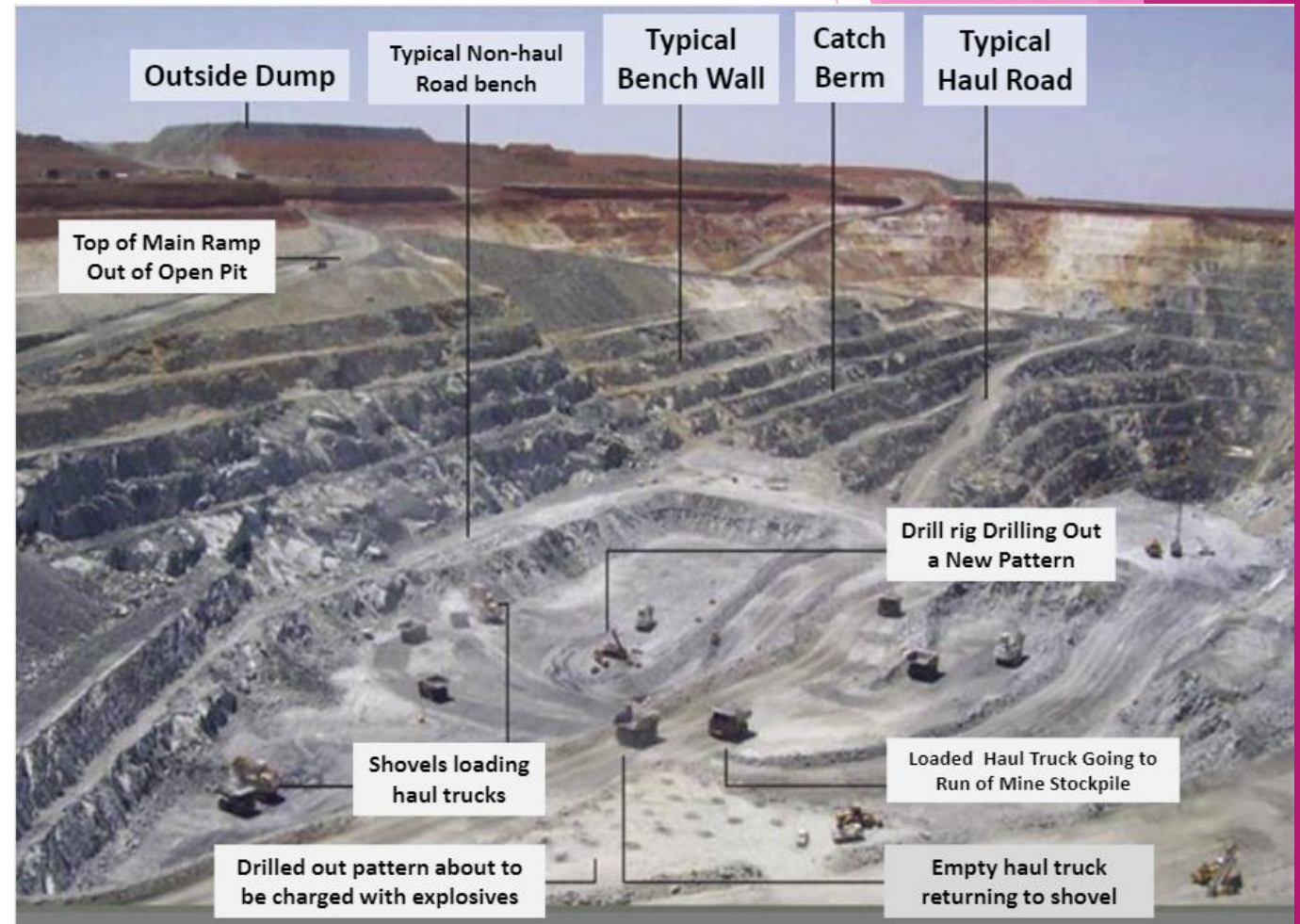
Pit Planning and Design

- ▶ Pit design and reserve estimation,
- ▶ Development of economic block model,
- ▶ cut-off grade and its estimation;
- ▶ Ultimate pit configuration and its determination -
 - ▶ Hand method,
 - ▶ Floating cone technique,
 - ▶ Lerchs-Grossmann algorithm, and
 - ▶ Computer assisted hand method.

Pit design and reserve estimation

The following are the key parameters affecting the pit design:

- ▶ Topography,
- ▶ Geology,
- ▶ Grade,
- ▶ Localization of the mineralization,
- ▶ Extent of the deposit,
- ▶ Property boundaries,
- ▶ Production rates,
- ▶ Road grades,
- ▶ Mining costs,
- ▶ Processing costs,
- ▶ Metal recovery,
- ▶ Marketing considerations,
- ▶ Bench height,
- ▶ Pit slopes,
- ▶ Cutoff grade,
- ▶ Strip Ratios (SR).



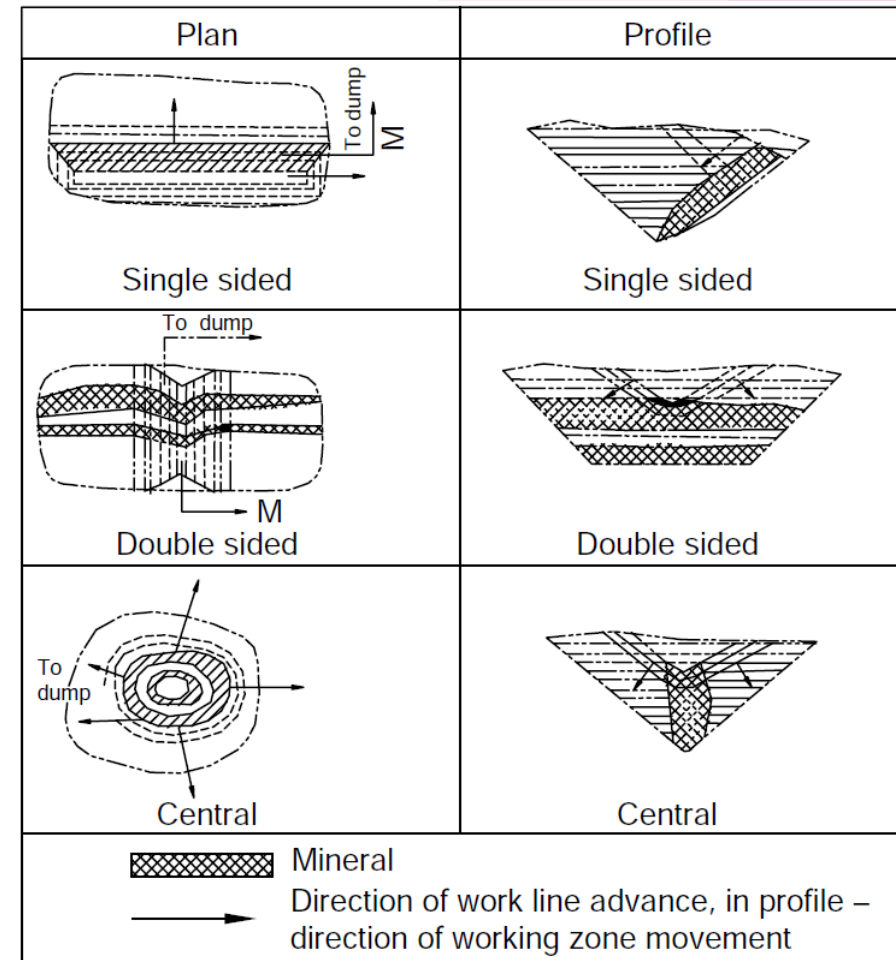
The end-purpose of your pit design will likely be

- ▶ Determining Ore Reserves
- ▶ Inputting into a schedule for Life-of-Mine planning
- ▶ To provide the guidance for the excavation of the pit to be detailed and laid out by the short-term design engineers.

As such your design needs to focus on

- ▶ Operational Efficiency (trucking and digging, and maybe drilling)
- ▶ Cost Minimization / Value Maximization (less waste, more ore).
- ▶ Schedule flexibility (is it practical to schedule and maintain productivity)
- ▶ Safety (don't build hazards and risks into the design!)

- ▶ **Open pit mining:** An open pit mine is the mine to exploit the deposits which are outcropping to the surface or those which are confined (limited) to a shallow (low) depth, and the waste rock lying above (overburden) is removed and transported away from the place of their deposition.
- ▶ **Open cast mining:** Opencast is also a surface mine to mine out the flat deposits but the overburden is backfilled in the worked-out area is removed by casting into mined-out areas, and mineral is excavated in consecutive operations.
- ▶ **Quarrying,** The term quarrying, of course, is very loosely applied to any of the surface mining operations but it should be confined to a surface mining method to mine out the dimensional stones such as slate, marble, granite etc. Quarrying is a highly specialized small-scale method, slow and the costliest of all mining methods. Only square set stopping method is as much expensive as quarrying.
- ▶ **Auger mining.** Auger mining is a surface mining technique used to recover additional coal from a seam located behind a highwall produced either by stripping or open-pit mining. Augering is employed to recover coal from the highwall at the pit limit.

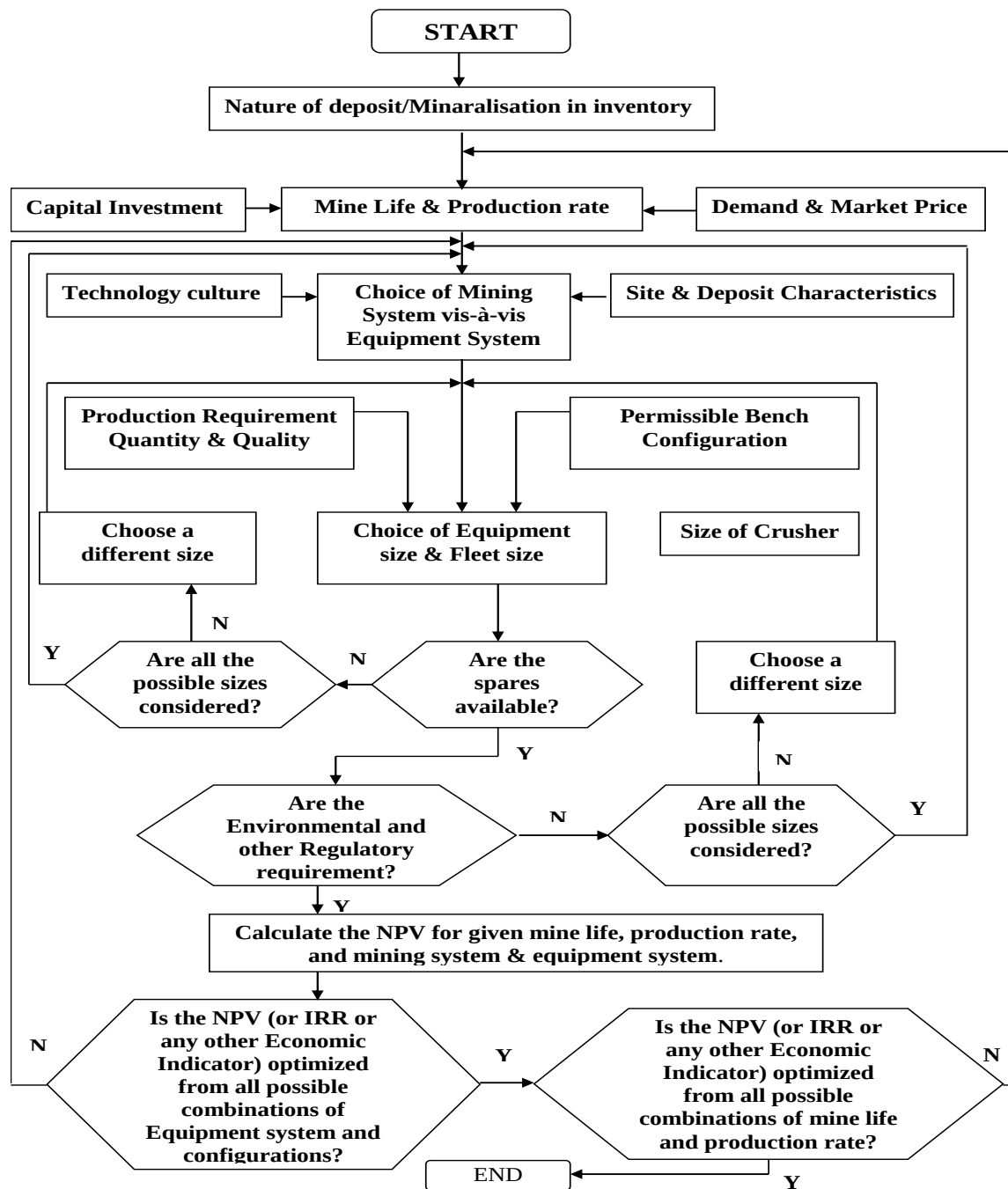


Planning phase involves three stages

- ▶ **Conceptual study** In this stage, the project idea is transformed into a broad investment proposition. It includes scope of work, definition and cost estimate. Capital and operating costs of the project are estimated using historical data. It is a work of one or two engineers.
- ▶ **Preliminary or pre-feasibility study** The main objective of a pre-feasibility study is to determine whether the project concept justifies the detailed feasibility study.
- ▶ **Feasibility study** It provides a definitive technical, financial and environmental base for an investment decision. It identifies production capacity, technology, investment, production costs, sales revenues and returns on investment.

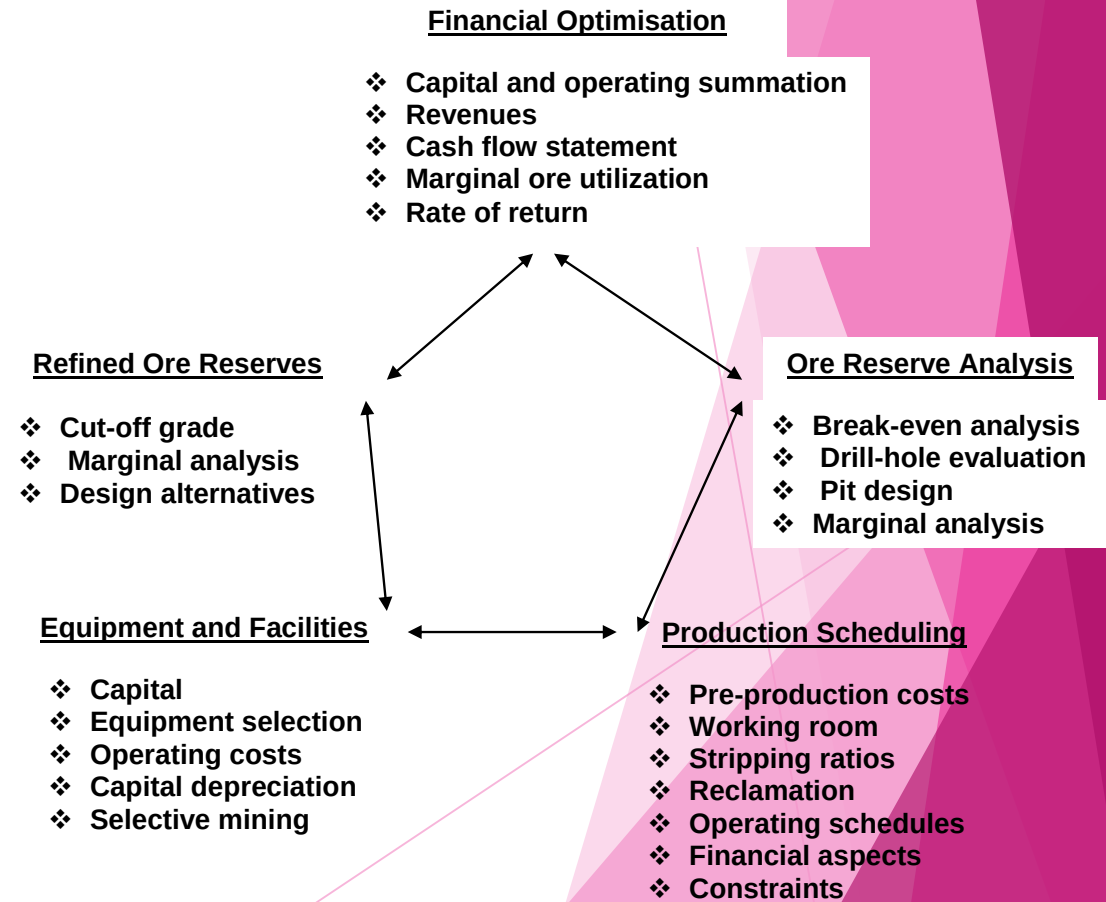
MINE PLANNING AND DESIGN STEPS

- ▶ A long-term price forecast and maximum practical marketable volume per year is Established.
- ▶ A geologic reserve assessment is made. A grade versus tonnage curve for the contiguous portion of the ore body including overburden / waste rock is developed. A typical cut-off grade for the commodity (gold, copper, iron, etc.), type of deposit (deep, shallow, etc.), and likely mining/processing system is selected.
- ▶ Block net values and costs for both ore and waste are developed using mining and milling rates ranging above and below those determined in No.2, incorporating capital costs and return consideration in the ore and waste block evaluations as feasible. Because timing and various inter-block interactions cannot be directly addressed, these will be only approximate.
- ▶ Sets of pit ultimate design are developed flexing (varying) prices and/or milling/mining rates and costs to get logical concentric nests of pits. (A block value of 0 to be used as cutoff).
- ▶ Short-range plans within each ultimate pit are developed using average operating slopes. Holding costs and rates constant, the price is flexed starting above the price of ultimate pit and work down in increments. This approach should mine the best ore/waste combinations first. A set of short-range mining segments will be generated in this process and thereby fix the feasible mine geometry options for a given ultimate pit.
- ▶ Production, revenue, operating and capital cost schedules over time are developed for each short-range segment and over full mine life from development through closure. These data will not normally be derived from the block data directly but they must be related for consistency.
- ▶ Value of each short-range segment is then optimized by flexing cut off criteria and mining/ore processing rate being careful not to exceed the marketable maximum volume, using NPV methods. The production and economic schedules in No.6 will be iterated to accomplish this. The choice of NPV discount rate will impact the results of the total evaluation process significantly.
- ▶ The NPV of the total pit plan segment sequence is evaluated to confirm optimum.



MINE PLANNING THROUGH CIRCULAR ANALYSIS FOR SURFACE MINES

The iterative nature of the planning and design process has been very aptly termed CIRCULAR ANALYSIS by Dohm (1979). The process and the included components are simply and rather elegantly presented in the following figure.



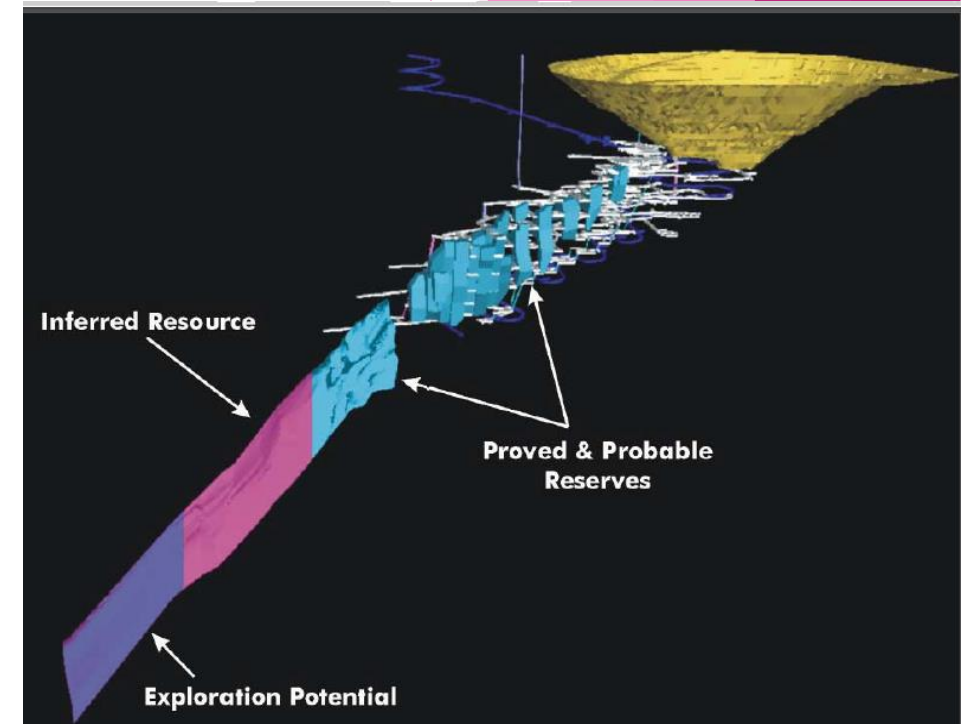
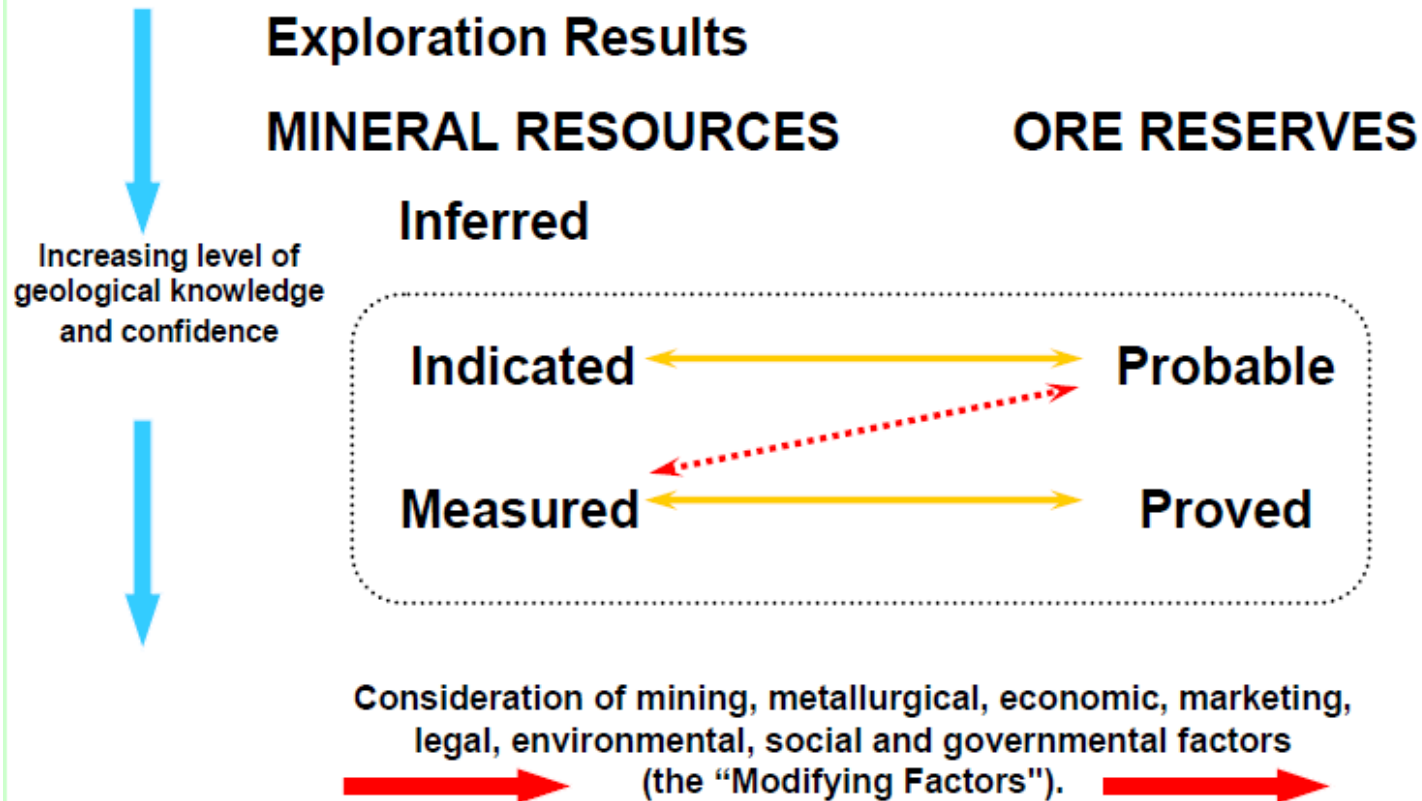
What is a Mineral Resource?

- ▶ *A Mineral Resource is an estimate of tonnage and grade for a mineralized body, based on sampling of that body*
- ▶ The estimate represents a realistic inventory that, under assumed and justifiable technical and economic conditions, might, in whole or in part, become economically extractable
- ▶ Portions of a deposit that do not have reasonable prospects for eventual economic extraction are **NOT Mineral Resources**

What is an Ore (Mineral) Reserve?

- ▶ An *Ore Reserve* is an *estimate of the tonnage* and grade that is expected to be delivered to the mill or treatment plant
- ▶ It is the economically mineable part of a Mineral Resource
- ▶ Realistically assumed Modifying Factors (mining, metallurgical, economic, marketing, legal, environmental, social and governmental factors) must be taken into consideration

Relationship between Mineral Resources and Ore Reserves



Mineral Resource Estimation

Requirements for estimating Mineral Resources:

- ▶ Confident geological interpretation
- ▶ High quality, representative samples and assays
- ▶ Application of appropriate estimation technique

This comes from:

- ▶ Mapping and sampling the deposit
- ▶ Ensuring the highest standards of sampling and assaying integrity
- ▶ Employing experienced, qualified professionals (“Competent Persons”)

Resource Estimation Techniques

- ▶ Methods for resource estimation or modeling are generally divided into
 - (1) traditional, **geometric methods that are done manually on plans or sections and**
 - (2) interpolation methods such as inverse-distance-weighting and kriging that require the use of a computer.
- ▶ These estimates are classified and reported in accordance with one of the accepted international reporting standards (e.g. UNFC, JORC, SAMREC)

Current Mineral Resource/Reserve Reporting Standards

- JORC (joint ore reserve committee) Code (Australasia) - translations into South American Spanish, Portuguese, Mandarin, Japanese and Russian (in progress)
- UNECE Framework Classification (International) - incorporates JORC style definitions for market-related reporting, but is broader in scope, covering government inventory reporting
- SAMREC Code (South Africa)
- CIM Standards (Canada, in NI 43-101)
- The Reporting Code (UK/Western Europe)
- SME Guidelines (USA)
- Chilean and Peruvian Codes
- CRIRSCO definitions and reporting standard template (International - in progress)

- ▶ The cut-off grade is the minimum ore grade that can be mined at a profit under economic conditions existing at a particular point in time. The cut-off grade can vary with time due to changes in such factors as commodity prices, operating costs, and taxes. The cut-off grade used for any reserve calculation should always be stated.

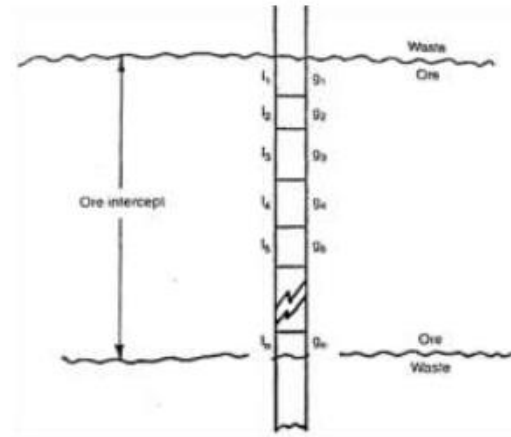


Fig. Ore intercept compositing

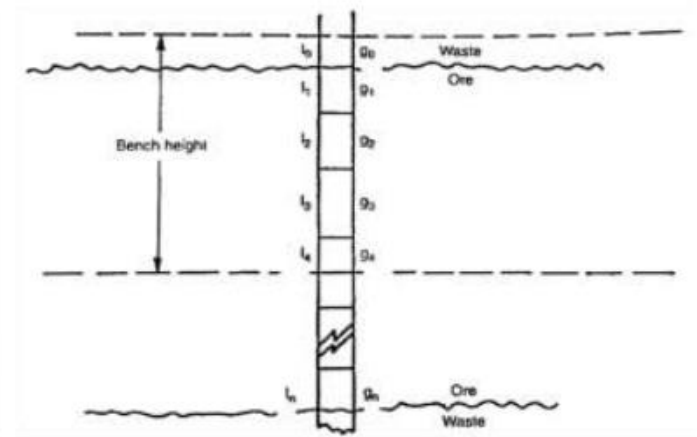


Fig. Bench height compositing

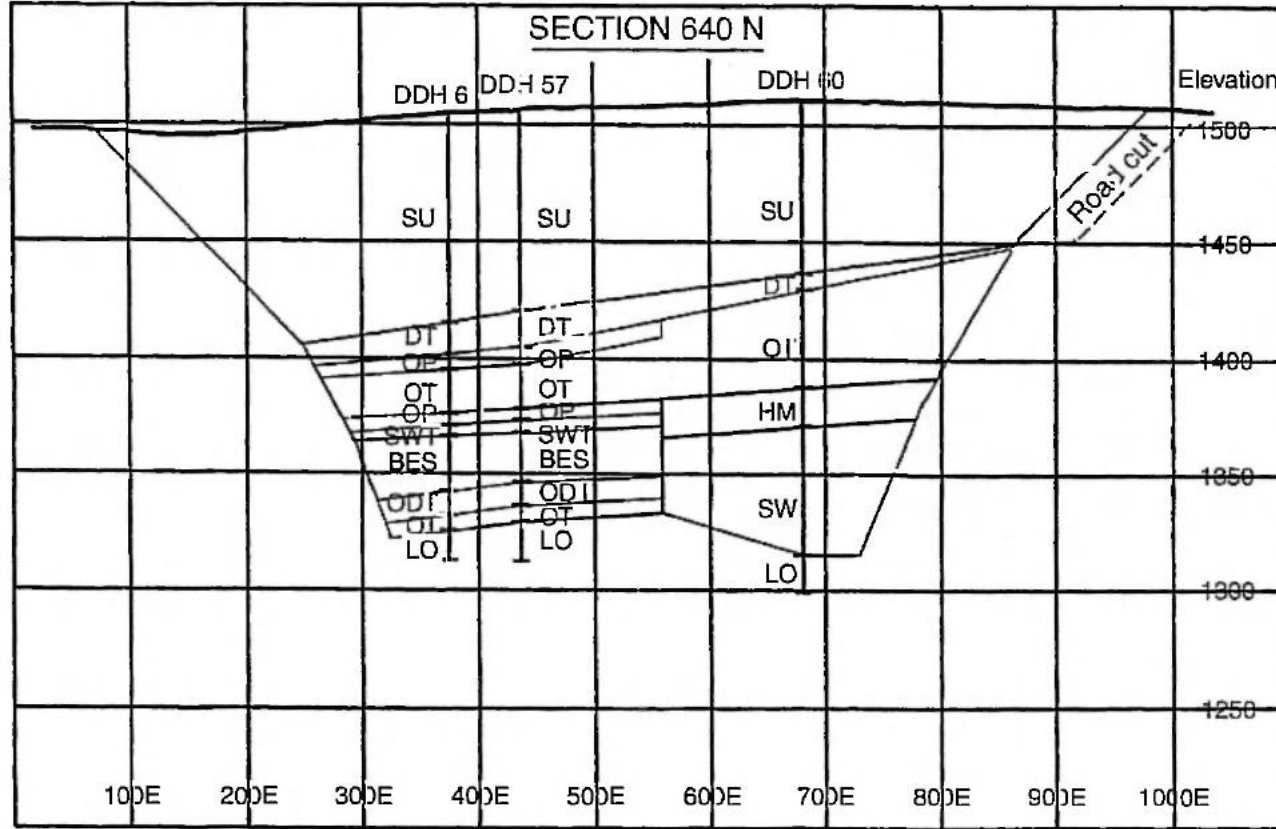
Grade Calculation

- ▶ Weighted average grade One of the most frequent calculations are weightings, e.g. for the calculation of the • average grade of a drill hole from assay intervals of different lengths or • average grade of a deposit from the combined grades of individual, unequal blocks.

Given		Calculated
Length	Grade	Length x Grade
l_1	g_1	$l_1 g_1$
l_2	g_2	$l_2 g_2$
l_3	g_3	$l_3 g_3$
l_n	g_n	$l_n g_n$
$\sum l_i$	$\bar{g}$	$\sum l_i g_i$

$$\bar{g} = \frac{l_1 \cdot g_1 + l_2 \cdot g_2 + \dots + l_n \cdot g_n}{l_1 + l_2 + \dots + l_n} \Rightarrow \bar{g} = \frac{\sum l_i g_i}{\sum L_i}$$

METHOD OF VERTICAL SECTIONS



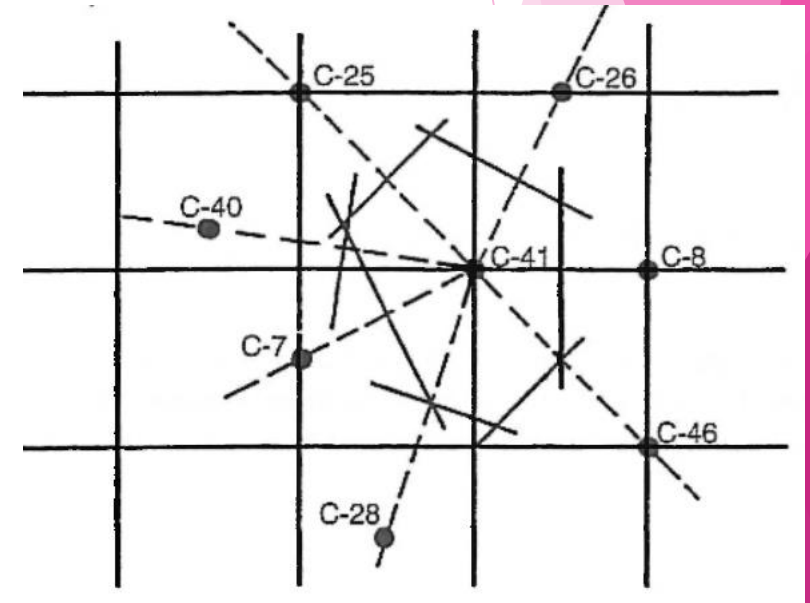
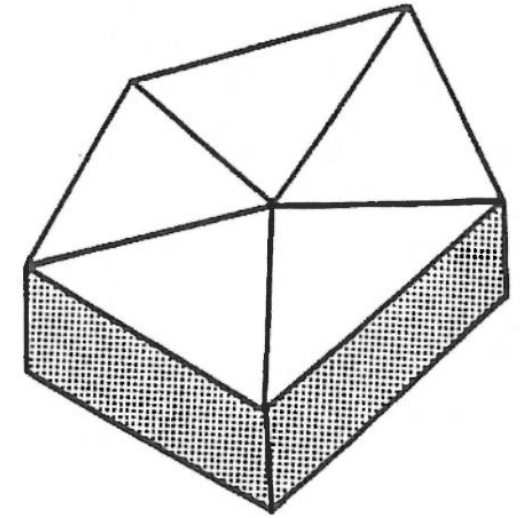
1. Side completion for sections.
2. Development of a final pit outline including pit ends.
3. Determination of tons and average grade for a section.
4. Determination of tons and average grade for the pit.

THE METHOD OF HORIZONTAL SECTIONS

- In the **triangular method**, each hole is taken to be at one corner of a triangle. If the triangular solid formed is of constant thickness t , its volume is just equal to the plan area A times this thickness. To obtain tons, the appropriate tonnage factor is applied. The average grade g is given by

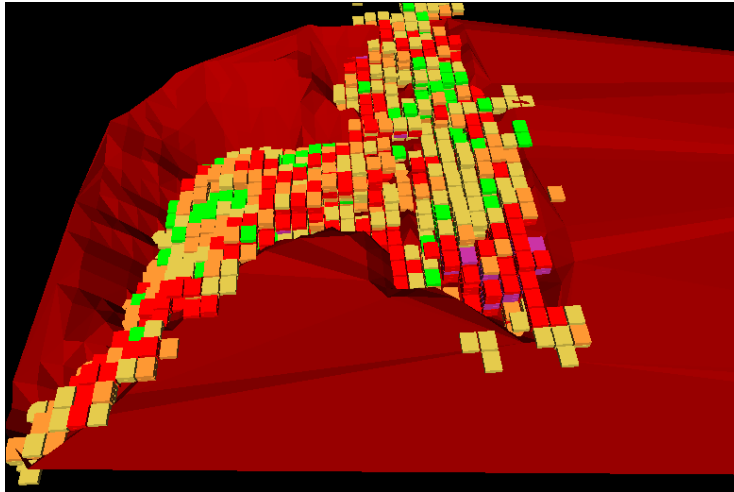
$$\bar{g} = \frac{g_1 + g_2 + g_3}{3}$$

In the **polygonal method**, each drill hole lies within the center of a polygon. The polygon is constructed such that its boundary is always equidistant from the nearest neighboring hole. Within the polygon, the grade is assumed constant and equal to that of the hole it includes. The thickness of the polygon is also constant and equal to the ore intercept/bench thickness



Block modelling

Basic to application of computer techniques for grade and tonnage estimation is the visualization of the deposit as a collection of blocks



Constant distance weighting techniques

inverse distance weighting technique.

The formula used to calculate this can be written as

$$g = \frac{\sum_{i=1}^n \frac{g_i}{d_i}}{\sum_{i=1}^n \frac{1}{d_i}}$$

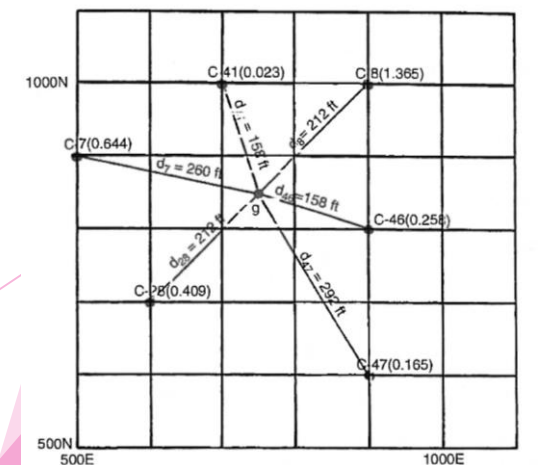
where g_i is the given grade at distance d_i away from the desired point.

Inversely with the square of the distance

$$g = \frac{\sum_{i=1}^n \frac{g_i}{d_i^2}}{\sum_{i=1}^n \frac{1}{d_i^2}}$$

If one were to select a different power for d , the results would change. The general formula is

$$g = \frac{\sum_{i=1}^n \frac{g_i}{d_i^m}}{\sum_{i=1}^n \frac{1}{d_i^m}}$$



- **Kriging** is the geostatistical estimation method developed to provide the “best linear, unbiased estimate” for grade based on a least squares minimization of the error of estimation, or kriging error.

1. The average of the estimates should not be systematically higher or lower than the true value; this is established mathematically by setting the sum of weights equal to zero, that is,

$$\sum w_i = 1 \quad (5.6.13)$$

2. The error of estimation σ_K^2 , which is expressed as variance $(G - G^*)$, or

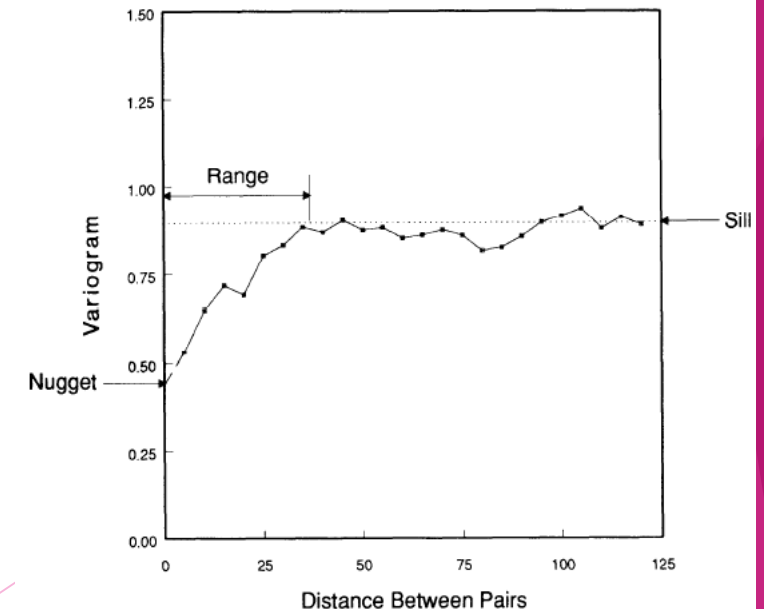
$$\sigma_K^2 = \sigma_{B,D}^2 - 2 \sum w_i \sigma_{B,X_i} + \sum \sum w_i w_j \sigma_{X_i X_j} \quad (5.6.14)$$

is minimized, where G is the actual grade, G^* is the estimated grade, $\sigma_{B,D}^2$ is the variance of blocks the size being estimated in the deposit, the σ_{B,X_i} are the covariances between each sample and the block being estimated, and the $\sigma_{X_i X_j}$ are the covariances between the individual samples.

The **variogram** is the fundamental tool used by the geostatistician and geologist to measure spatial continuity of grade data. The variogram is a graph of the average variability between samples vs. the distance between samples. A variogram is computed by averaging the squared differences between pairs of samples that are a given distance apart as follows

$$\begin{aligned} \gamma(h) &= \text{variogram for distance } h \\ &= \frac{1}{2N} \sum (g_i - g_{i+h})^2, i = 1, 2, 3 \dots N \end{aligned}$$

where N is the number of pairs at distance h , and h is the distance between the samples



Main Factors affecting Mineral Resource/ Reserve Estimates

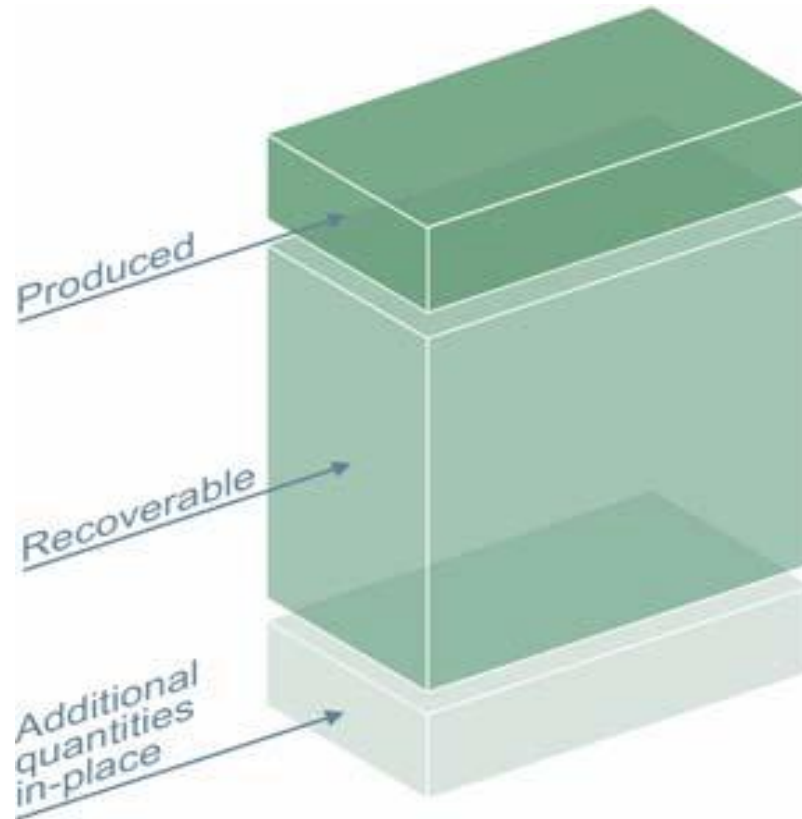
- ▶ Reliability of geological interpretation
- ▶ Amount, distribution and quality of resource data
- ▶ In “nuggety” deposits, treatment of very high grades
- ▶ Assumptions regarding mining and treatment methods
- ▶ Assumptions regarding commodity prices and exchange rates
- ▶ Experience and judgment of Competent Person

UNFC-Total Initial Resources

The total resources initially in-place of naturally occurring energy and mineral resources, are described in terms of:

- ▶ Produced quantities,
- ▶ Remaining recoverable quantities, and
- ▶ Additional quantities remaining in-place.

The main focus of the UNFC is on remaining recoverable quantities.



Total initial in-place resources

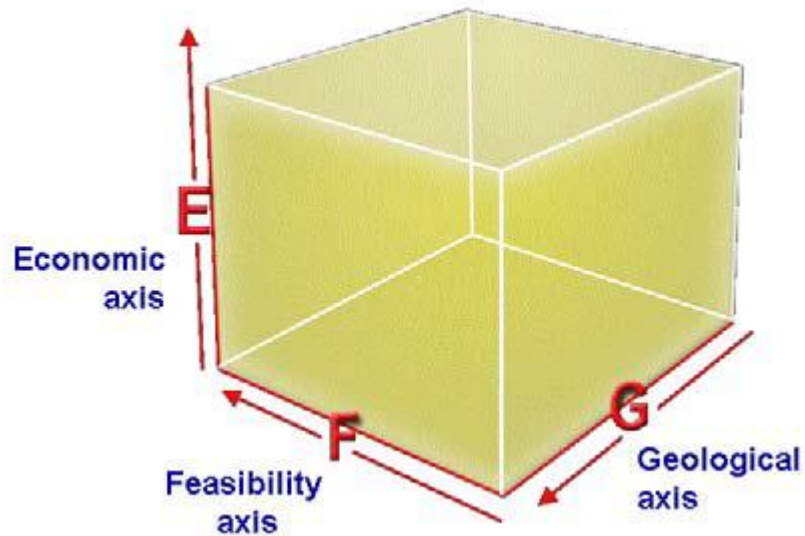
For non-renewable resources, the total resources initially in-place is constant. In inventories, material balance is therefore maintained. If any change appears, this must be explained by a re-evaluation.

Classifications

Total remaining resources are categorized using the three essential criteria affecting their recoverability:

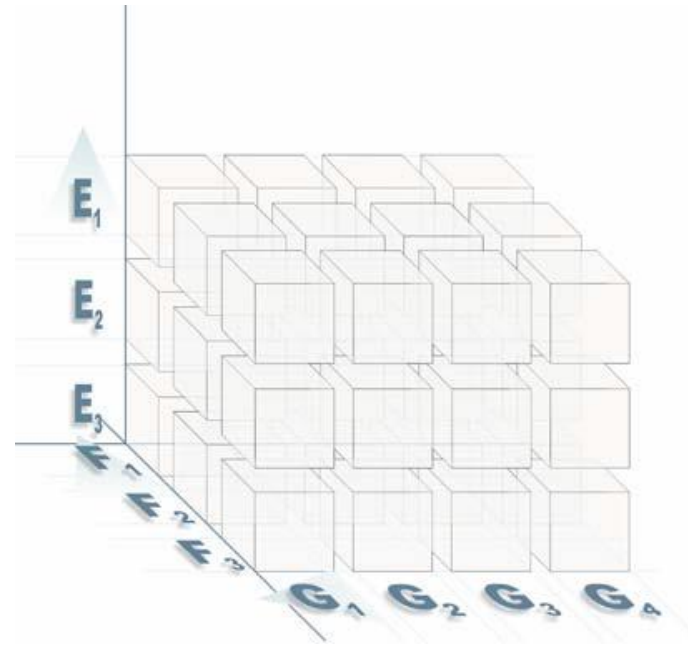
- ▶ Economic and commercial viability (E).
- ▶ Field project status and feasibility (F).
- ▶ Geological knowledge (G).

- ▶ Most of the existing resource classifications recognize these explicitly or implicitly. By making them explicit, the UNFC becomes a framework that allows for harmonization of existing classifications. The three criteria are easily visualized in three dimensions as shown in Figure



Principal elements of the UNFC

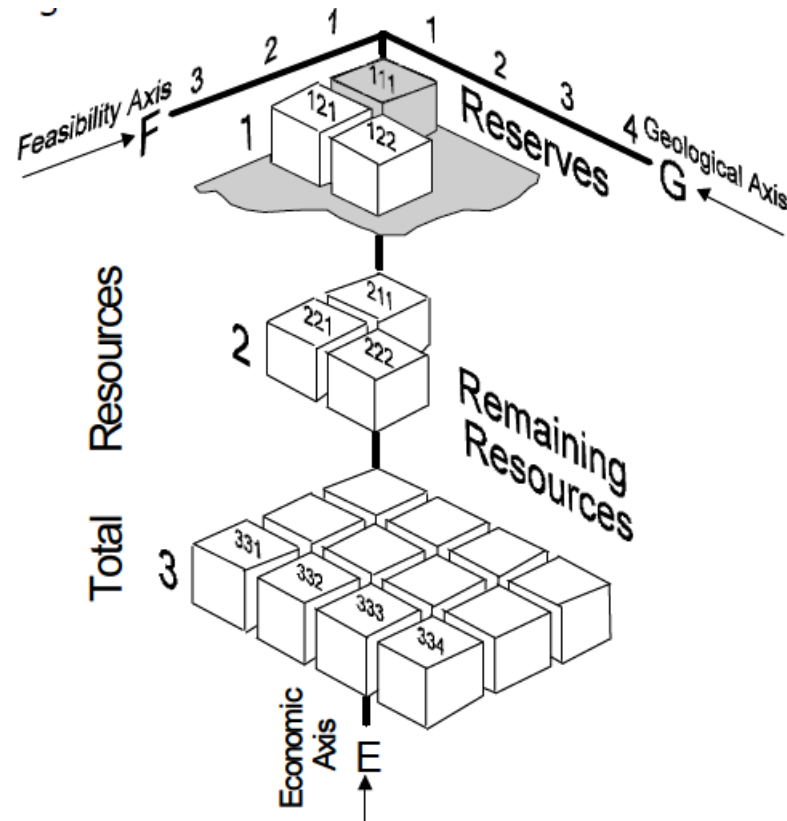
- ▶ Three main categories are used to describe economic and commercial viability,
- ▶ three to describe field project status and feasibility and
- ▶ four to describe the level of geological knowledge.



Further subdivision of the main categories is useful for special applications. Resource quantities are then grouped into classes that are defined by an E a F and a G category represented by the sub-cubes in Figure . A class of quantities may be a single sub-cube, i.e. 111, or a collection of sub-cubes. Total resources are an example of such a class where all sub-cubes are included in the class.

The three dimensions of categorization are represented by the edges of a cube. The digits are quoted in the order EFG firstly because the alphabetical order is easy to memorize, and secondly because the first digit refers to the economic viability, which is of decisive interest to producers, investors and host countries. Numbers are used to designate the different classes. Number 1, in accordance with the usual perception that the first is the best, refers to the highest degree of economic viability on the E axis, the most advanced project status on the F axis and the highest quality assessment on the G axis. The use of categories is different for fluids and for solids. This is primarily due to the fact that fluids may flow in a reservoir, irrespective of the level of geological knowledge. In the case of solids, recovery will normally be restricted to rock bodies that have been reliably assessed.

Codification



Due to variation between terminologies in different systems and languages, it is recommended to use only three-digit numeric codes for individual categories, so that they will be universally understood. For this to be possible, the sequence is always fixed, so that the quantity characterized as E1;F1;G1 may be written in number form as 111, independent of languages. In practice, only a limited number of combinations (classes) are valid. To illustrate, the UNFC for coal, uranium and other solid minerals, shown in Figure

The UNFC in matrix form applied to coal, uranium and other solid minerals

UN International Framework		Detailed Exploration	General Exploration	Prospecting	Reconnaissance
	National System				
Feasibility Study and/or Mining Report		1 (111)	usually		
		2 (211)			
Prefeasibility Study		1 (121) + (122)	not relevant		
		2 (221) + (222)			
Geological Study*)		3 (331)	3 (332)	3 (333)	3 (334)

Economic Viability Categories:

1: economic
2: potentially economic

3: intrinsically economic (economic to potentially economic)

cut-off grade and its estimation

- The cut-off grade is the minimum ore grade that can be mined at a profit under economic conditions existing at a particular point in time.

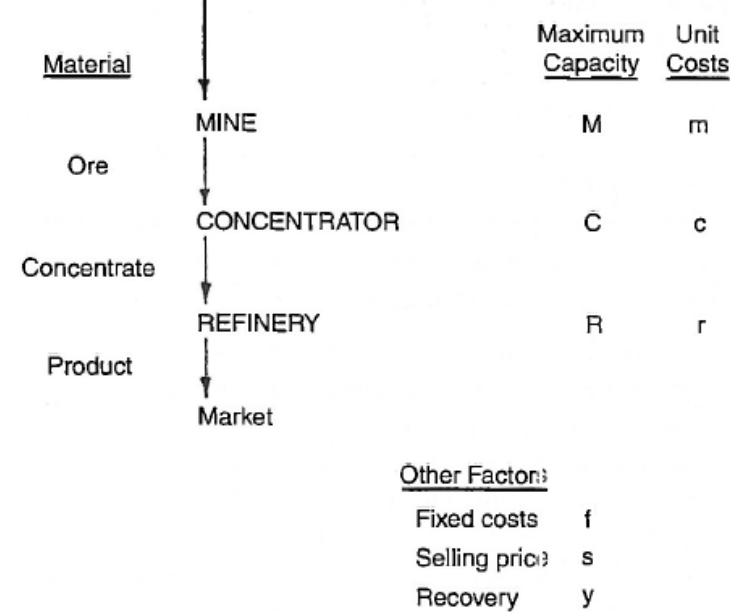


Figure 6.21. The model described by the Lane algorithm (Lane, 1964).

LANE'S ALGORITHM

cut-off grade is the criterion normally used in mining to discriminate between ore and waste in the body of a deposit. Waste may either be left in place or sent to waste dumps. Ore is sent to the treatment plant for further processing and eventual sale.

The choice of cutoff grade can directly affect profits. This chapter will examine the principles which determine the best choice of a cutoff grade under different circumstances.

A mining operation is considered to consist of three stages:

- mining,
- concentrating, and
- refining.

Each stage has its own associated costs and a limiting capacity. The operation as a whole will incur continuing fixed costs. The three most important economic criteria which can be applied are:

Case I: Maximum present value.

Case II: Maximum total profits.

Case III: Maximum immediate profit.

Quantities: T is the length of the production period being considered (for example 1 year); Q is the quantity of material to be mined, Q_c is the quantity of ore sent to the concentrator and Q_r is the amount of product actually produced over this production period.

The basic equations

Using the definitions given in the preceding section, the basic equations can be developed. The total costs T_c are

$$T_c = mQ_m + cQ_c + rQ_r + fT$$

Since the revenue $\bar{R}$ is

$$\bar{R} \equiv sQ_r$$

the profit P is given by

$$P = R - T_c \equiv sQ_r - (mQ_m + cQ_c + rQ_r + fT)$$

Combining terms, yields

$$\bar{P} \equiv (s - r)Q_r - cQ_c - mQ_m - fT$$

This is the basic profit expression. It can be used to calculate the profit from the next Q_m of material mined.

Cutoff grade for maximum profit

Step 1. Determination of the economic cutoff grade – one operation constraining the total capacity

As indicated, the basic profit expression (6.10) is

$$\bar{P} = (s - r)Q_r - cQ_c - mQ_m - fT$$

Calculate cutoff grade assuming that the mining rate is the governing constraint. If the mining capacity M is the applicable constraint, then the time needed to mine material Q_m is

$$T_m = \frac{Q_m}{M} \quad (6.11)$$

Equation (6.10) becomes

$$P = (s - r)Q_r - cQ_c - \left(m + \frac{f}{M}\right) Q_m \quad (6.12)$$

To find the grade which maximizes the profit under this constraint one first takes the derivative of (6.12) with respect to g .

$$\frac{dP}{dg} = (s - r) \frac{dQ_r}{dg} - c \frac{dQ_c}{dg} - \left(n + \frac{f}{M}\right) \frac{dQ_m}{dg} \quad (6.13)$$

However the quantity to be mined is independent of the grade, hence

$$\frac{dQ_m}{dg} = 0 \quad (6.14)$$

Equation (6.13) becomes

$$\frac{dP}{dg} = (s - r) \frac{dQ_r}{dg} - c \frac{dQ_c}{dg} \quad (6.15)$$

The quantity refined Q_r is related to that sent by the mine for concentration Q_c by

$$Q_r = \bar{g}yQ_c \quad (6.16)$$

where $\bar{g}$ is the average grade sent for concentration, and y is the recovery.

Taking the derivative of Q_r with respect to grade one finds that

$$\frac{dQ_r}{dg} = \bar{g}y \frac{dQ_c}{dg} \quad (6.17)$$

Substituting Equation (6.17) into (6.15) yields

$$\frac{dP}{dg} = [(s - r)\bar{g}y - c] \frac{dQ_c}{dg} \quad (6.18)$$

The lowest acceptable value of $\bar{g}$ is that which makes

$$\frac{dP}{dg} = 0$$

Thus the cutoff grade g_m based upon mining constraints is the value of $\bar{g}$ which makes

$$(s - r)\bar{g}y - c = 0$$

Thus

$$g_m = \bar{g} = \frac{c}{y(s - r)} \quad (6.19)$$

- *Calculate cutoff grade assuming that the concentrating rate is the governing constraint.* If the concentrator capacity C is the controlling factor in the system, then the time required to mine and process a Q_c block of material (considering that mining continues simultaneously with processing) is

$$T_c = \frac{Q_c}{C}$$

Substituting Equation (6.20) into (6.10) gives

$$P = (s - r)Q_r - cQ_c - mQ_m - f \frac{Q_c}{C}$$

Rearranging terms one finds that

$$P = (s - r)Q_r - \left(c + \frac{f}{C}\right) Q_c - mQ_m$$

Differentiating with respect to g and setting the result equal to zero yields

$$\frac{dP}{dg} = (s - r) \frac{dQ_r}{dg} - \left(c + \frac{f}{C}\right) \frac{dQ_c}{dg} - m \frac{dQ_m}{dg} = 0$$

As before

$$\frac{dQ_m}{dg} = 0$$

$$\frac{dQ_r}{dQ_c} = gy$$

Thus

$$(s - r) \frac{dQ_r}{dg} - \left(c + \frac{f}{C}\right) \frac{dQ_c}{dg}$$

$$\frac{dQ_r}{dQ_c} = \frac{c + \frac{f}{C}}{s - r} = gy$$

The cutoff grade when the concentrator is the constraint is

$$g_c = \frac{c + \frac{f}{C}}{y(s - r)}$$

- Calculate cutoff grade assuming that the refining rate is the governing constraint. If the capacity of the refinery (or the ability to sell the product) is the controlling factor then the time is given by

$$T_r = \frac{Q_r}{R}$$

Substituting (6.23) into Equation (6.10) yields

$$P = (s - r)Q_r - \frac{fQ_r}{R} - cQ_c - mQ_m$$

or

$$P = \left(s - r - \frac{f}{R} \right) Q_r - cQ_c - mQ_m$$

Differentiating with respect to g and setting the result equal to zero gives

$$\frac{dP}{dg} = \left(s - r - \frac{f}{R} \right) \frac{dQ_r}{dg} - c \frac{dQ_c}{dg} - m \frac{dQ_m}{dg} = 0$$

Simplifying and rearranging gives

$$g_r = \frac{c}{\left(s - r - \frac{f}{R} \right) y}$$

Brief-Cutoff grade and Reserves

Cutoff grade could be defined as “any grade that for any specific reasons, is used to separate two courses of action, e.g. to mine or to dump”.

Where grade of the mineralized material is less than cutoff grade it is classified as waste and where it is equal to or above cutoff grade it is classified as ore.

The reasons for continuing interest in cutoff grades are obvious. Too high a grade can reduce the mineral recovered and possibly the life of the deposit (figs 12.3(a) to (d)). Too low a cutoff would reduce the average grade (and hence profit) below an acceptable level. In project evaluation it is important to determine a cutoff grade, which is normally set to achieve the financial objectives for the project.

Studies on cutoff grade theory may fall into two basic categories.

The fixed cutoff grade concept assumes a static cutoff for the life of the mine, while the variable cutoff grade concept assumes a dynamic cutoff maximizing the mine net present value.

Lane outlined three distinct stages in a mining operation: ore generation (mining), concentration (milling) and refining. He demonstrated that in establishing cutoff grades, consideration of costs, capacities, waste: ore ratios and average grade of different increments of ore of the orebody as well as the present values of annual cash flows are essential. For each stage, there is a grade at which the cost of extracting the recoverable metal equals the revenue from the metal. This is commonly known as *break-even grade*.

Constant price and cost

Simple formula $(s-r) y g = c$

Maximum total profit

$(s-r) y g = c + f/C$

Maximum present value

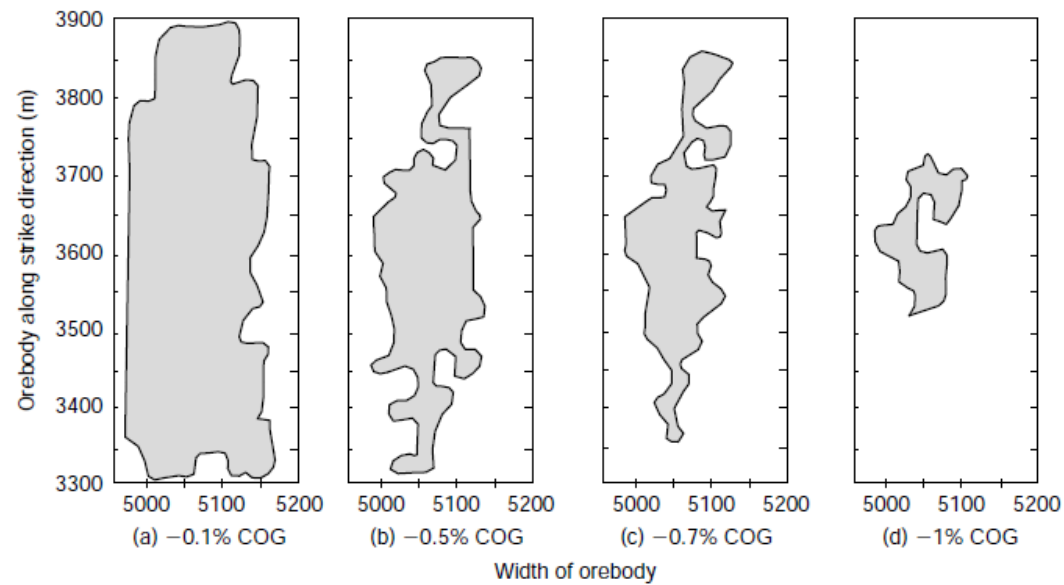
$(s-r) y g = c + (f + d \times v)/C$

Varying prices and costs

Maximum present value

$(s-r) y g = c + (f + d \times v + \delta v)/C$

Whereas: s – price of metal/mineral
 r – smelting and marketing costs
 y – recovery of mineral
 c – marginal (variable) processing costs
 f – annual fixed cost
 d – discount rate
 C – capacity: units of ore p.a.
 g – cutoff grade
 v – present value
 δv – the decline in present value over a year.



$$\text{Workable or Mineable Reserves, } Q_w = Q_g - N_w \quad (12.1a)$$

$$\text{Commercial Reserves, } Q_c = Q_w - O_L \quad (12.1b)$$

Whereas: Q_g – Geological Reserves.

Q_w – Mineable or workable reserves.

Q_c – Commercial reserves.

N_w – non-workable reserves.

O_L – Total operational losses = $L_1 + L_2 + L_3$

L_1 – losses in pillars, barriers, boundaries etc.

L_2 – losses due to poor breakage, blasted ore losses in worked out stopes, etc.

L_3 – losses due to abnormal adverse working conditions such as excessive ground pressure, heat, humidity, gases, water etc.

Mineral Inventory:¹ For any mineral deposit accountability of its reserves grade-wise is known as the mineral inventory assessment for that deposit. Thus, there is difference between mineral inventory and ore reserves. In ore reserves estimation the technique, method, equipment to mine out the deposit together with its cost of mining and selling price comes into picture, whereas, in mineral inventory estimation these considerations are not required to be made.

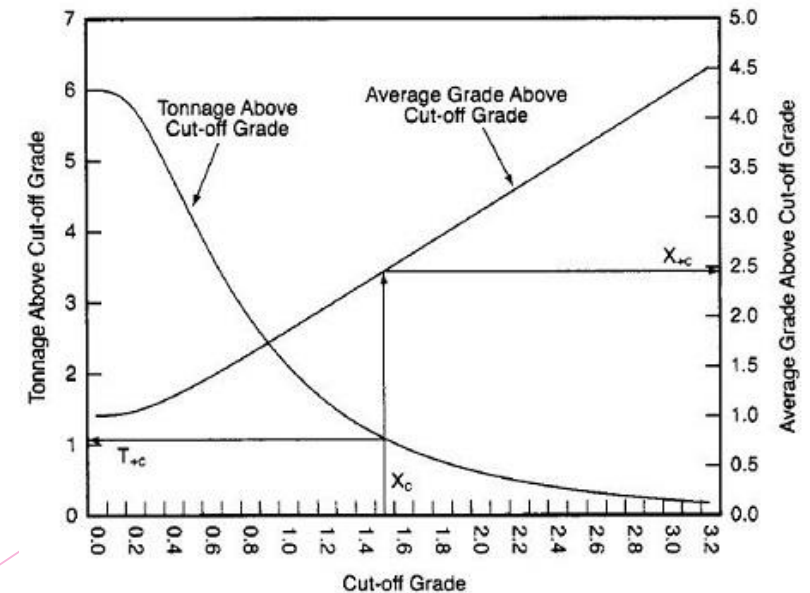
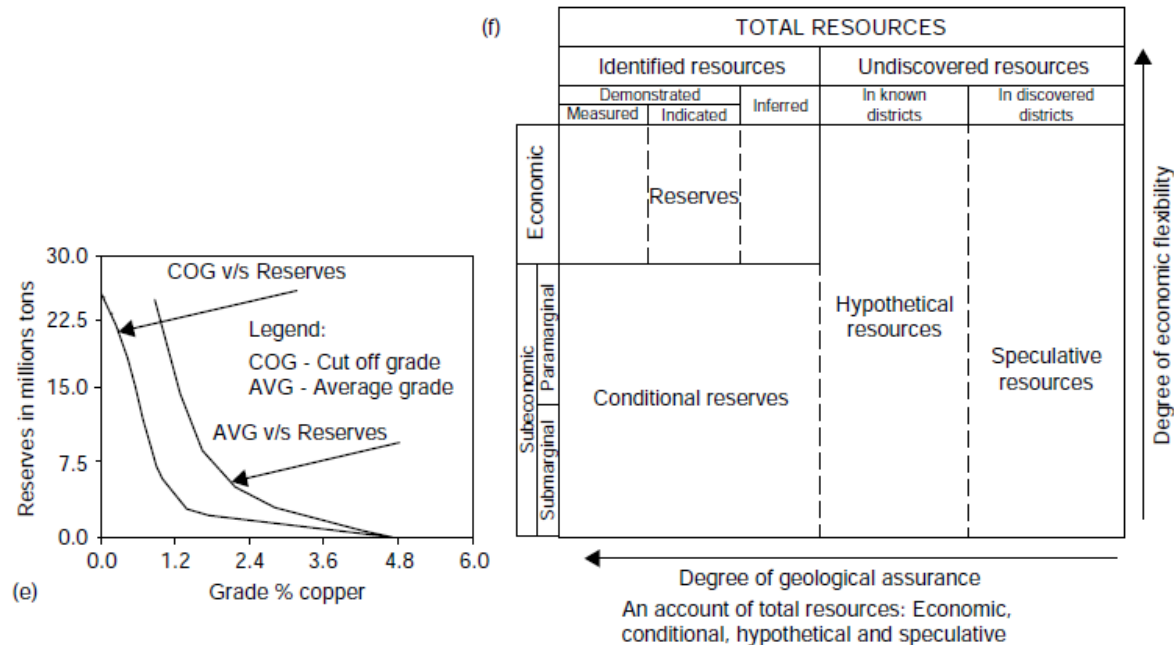
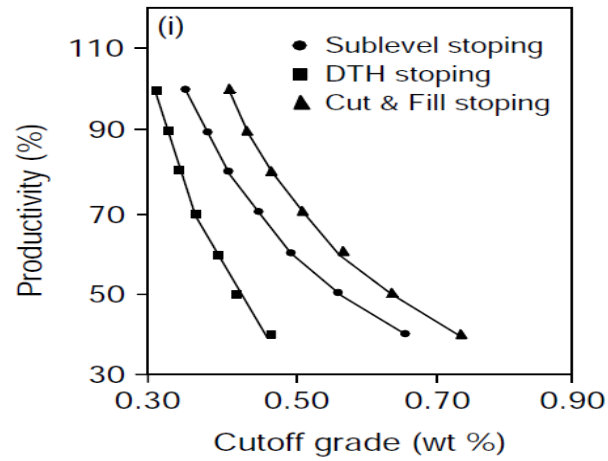
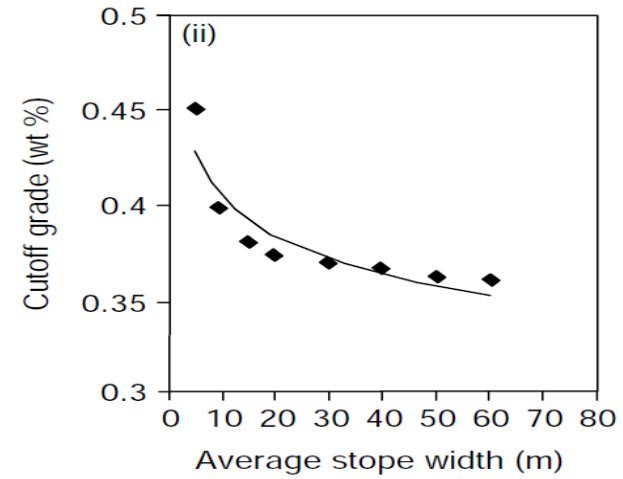


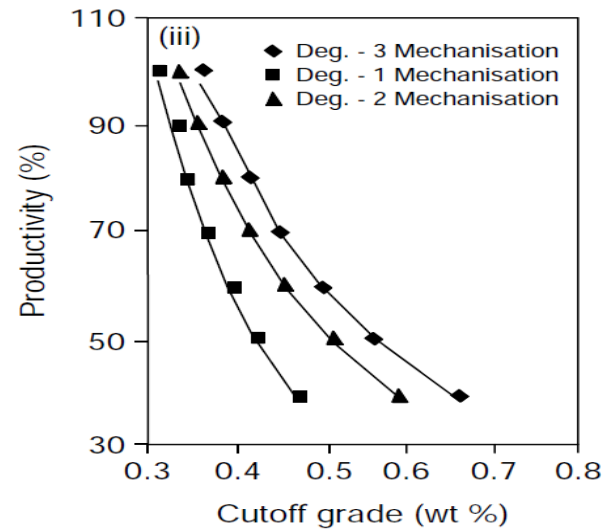
Figure 12.3 (a to d) : Influence of raising cutoff grade on orebody profile and reserves. (e): Grade – Tonnage curve. (f): An account of total resources.



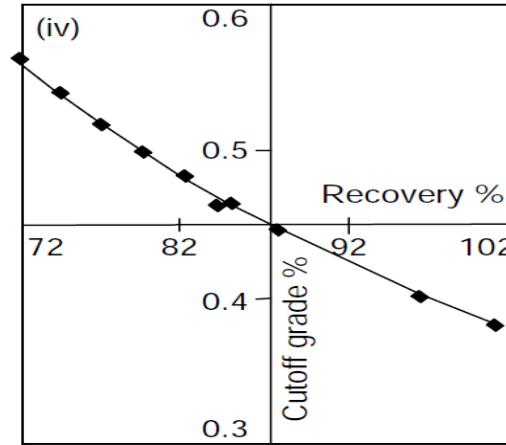
Influence of orebody thickness (stope width) on cutoff grades.



Influence of stoping methods at different levels of productivity.



Influence of mechanisation on cutoff grade.



Influence of overall process Recoveries on cutoff grades.

(a)

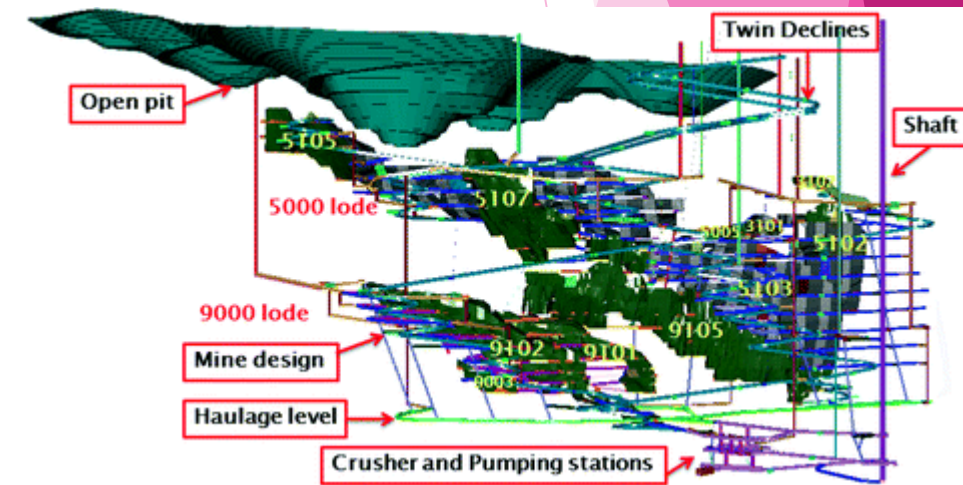
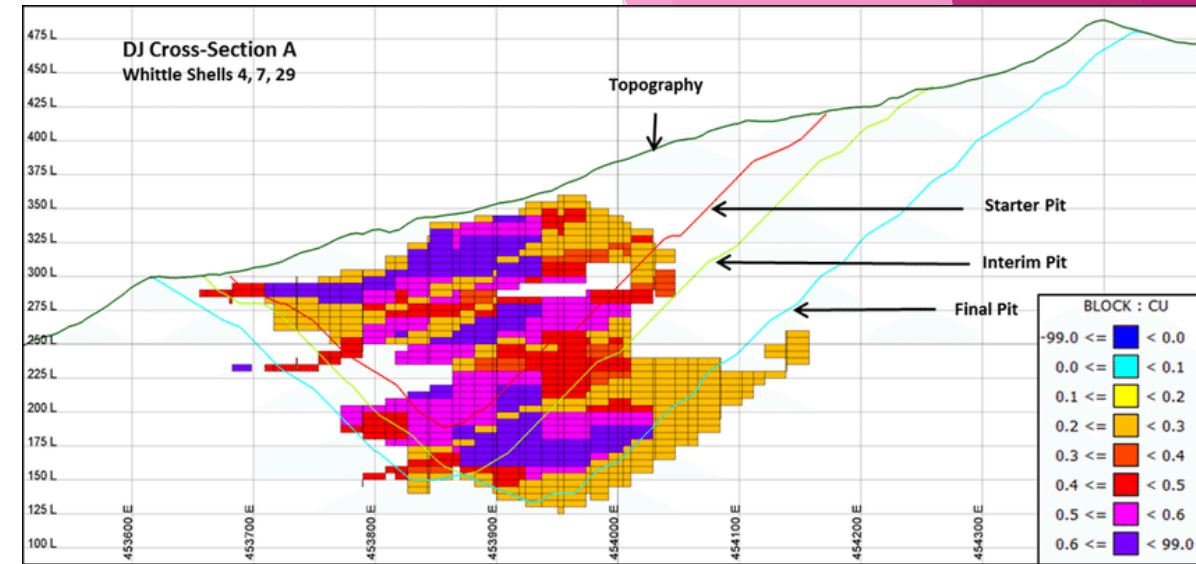
Parameter influencing cutoff grades.

► Ultimate pit configuration and its determination -

- Hand method,
- Floating cone technique,
- Lerchs-Grossmann algorithm, and
- Computer assisted hand method.

INTRODUCTION

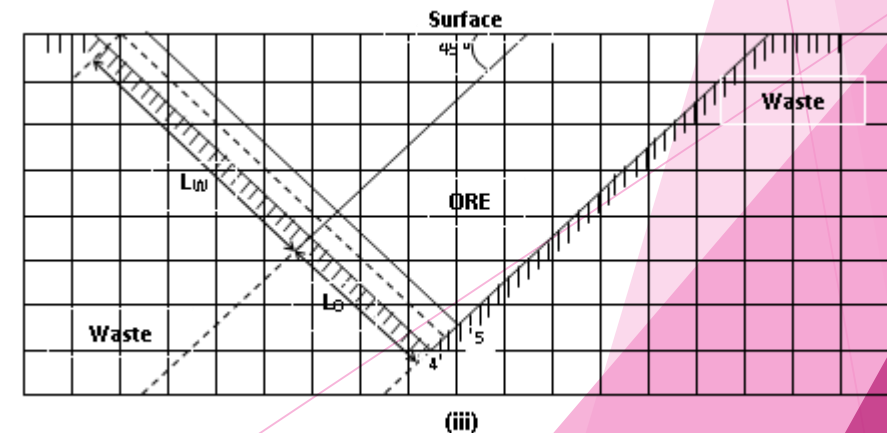
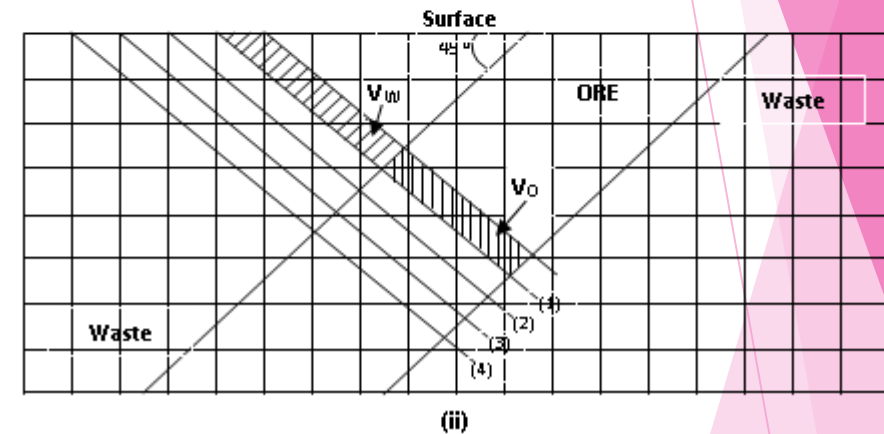
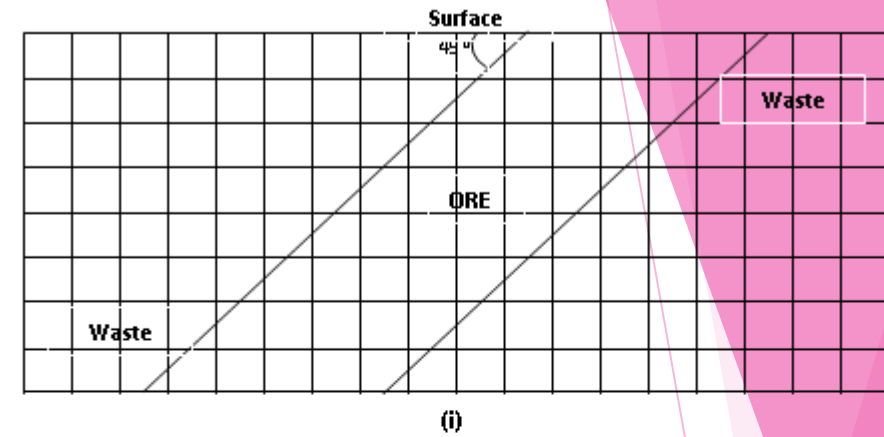
Once the mineral inventory is developed, the next step involves the development and superimposition of the pit onto the mineral inventory. The mineable material becomes that lying within the pit **BOUNDARIES**. The size and shape of the pit is a function of economic factors and design/production constraints. With an increase of selling price of mineral/ore, the pit would expand in size assuming all other factors remain constant and vice-e-versa. The pit configuration existing at the end of mining is called the 'final pit configuration' or the 'ultimate pit configuration (UPC)'. The UPC may be determined by different methods - hand methods, computer methods and computer assisted hand methods.



Hand Methods

In the hand methods, there are following four basic steps involved in determining the UPC.

- ▶ A slice is selected.
- ▶ The contained value is compared with the costs.
- ▶ If the net value is positive, the pit can be expanded, and if it is negative, the pit contracts.
- ▶ The final pit position is where the net value of the slice is zero.



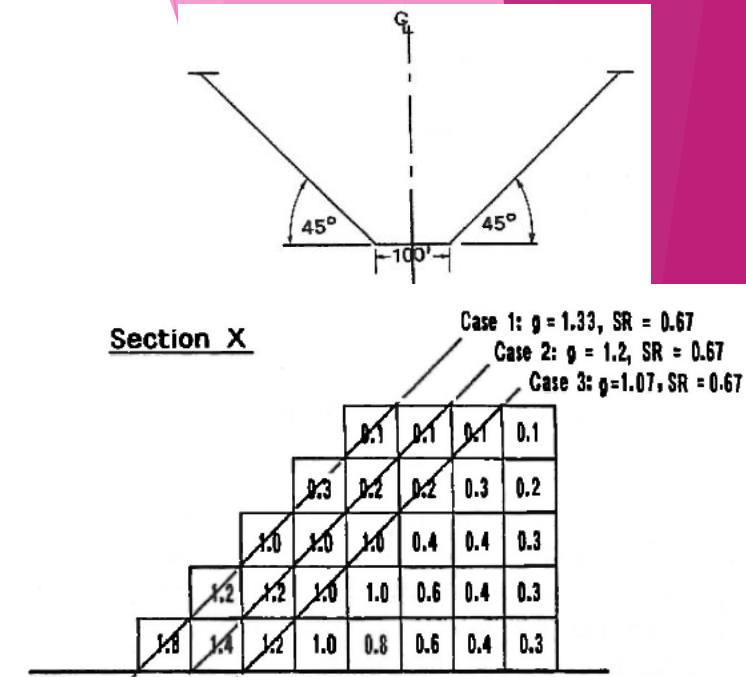
Floating Cone Method

In floating cone method for determination of UPC, the following steps are used.

- ▶ The cone is floated from left to right along the top row of blocks in the section. If there is a positive block, it is removed.
- ▶ After traversing the first row, the apex of the cone is moved to the second row. Starting from the left hand side it floats from left to right stopping when it encounters the first positive block. If the sum of all the blocks falling within the cone is positive (or zero), these blocks are removed (mined). If the sum is negative the blocks are left, and the cone floats to the next positive block on this row. The summing and mining or leaving process is repeated.
- ▶ This floating cone process moving from left to right and top to bottom of the section continues until no more blocks can be removed.
- ▶ The profitability for this section is found by summing the values of the blocks removed.
- ▶ The overall stripping ratio for the section can be determined from the numbers of positive and negative blocks to be mined.

The problems that may be encountered in using this method are:

1. Missing combination of profitable blocks
2. Extending the ultimate it beyond the optimal pit limit
3. Combination of problems 1 and 2.



-1	-1	-1	-1	-1	-1	-1
	-2	-2	-4	-2	-2	
		7	1	-3		

-1	-1	-1	-1	-1	-1	-1
	-2	-2	-2	-2	-2	
		10	-3	10		

(i)

-1	-1	-1	-1	-1
	5	-2	-2	
		5		

$$(ii)$$

-1	-1	-4	-1	-1
	5	-4	5	
		3		

(iii)

-1	-1	-1	-1	-1	+1	-1
	-2	-2	+4	-2	-2	
		+7	+1	-3		

Block
Value

Figure 5.40. First incremental cone – Example 1 (Barnes, 1982).

-1	-1	-1	-1	-1		-1
	-2	-2	+4	-2	-2	
		+7	+1	-3		

Block
Value

Figure 5.41. Second incremental cone – Example 1 (Barnes, 1982).

-1	-1					-1
	-2	-2		-2	-2	
		+7	+1	-3		

Block
Value

Figure 5.42. Third incremental cone – Example 1 (Barnes, 1982).

Lerchs-Grossmann 2-D Algorithm

In the Lerchs-Grossmann 2-D algorithm for determination of UPC, the following steps are used.

- ▶ Calculate cumulative profits for each column ($j = 1$ to J) of blocks starting from the top ($i = 1$) to the bottom ($i = I$) moving downward. Each vertical column of blocks is independent of the others.
- ▶ Add artificial row ($i = 0$) containing zeros and two artificial columns ($j = 0$ and $j = J+1$) containing zeros.
- ▶ Develop an overall cumulative sum moving laterally from left to right across the section. Beginning with the extreme top left hand block, the values of three blocks are examined: (i) one directly above and to the left, (ii) one on the left, and (iii) one directly below and to the left. Of the three, that block which when its value is added to the block in question yields the most positive selected. An arrow is drawn from the original block to that block. This sum is substituted for that originally assigned and becomes the value to be used for subsequent calculations.
- ▶ This process is continued, working down the first column, then down the next column to the right, until all blocks have been treated.
- ▶ The value of the top rightmost block is checked. If the value is zero or negative, then no profitable pit exist. If the value is positive, that is the value of the optimum pit.
- ▶ The blocks to be mined are identified by tracing back the arrow directions from right to left.

-2	-1	-1	0	0	0	0	-1	-1	-2
-2	0	0	1	2	1	1	1	0	-1
-1	0	1	2	2	3	3	2	1	-1
-1	-1	1	2	-1	2	1	1	0	-1

(i)

-2	-1	-1	0	0	0	0	-1	-1	-2
-4	-1	-1	1	2	1	1	0	-1	-3
-5	-1	0	3	4	4	4	2	-	-4
-6	-2	1	5	3	6	5	3	1	-5

(ii)

0	0	0	0	0	0	0	0	0	0	0	0
0	-2	-1	-1	0	0	0	0	-1	-1	-2	0
		-1	-1	1	2	1	1	0	-1		
			0	3	4	4	4	2			
				5	3	6	5				

(iii)

0	0	0	0	0	0	0	3	7	10	15	16
	-2	-1	-1	0	0	3	7	10	15	16	
		-3	-2	0	3	7	11	16	18		
			-3	1	6	10	16	19			
				2	5	12	17				

(iv)

-4	-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
	-4	-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
		-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
			-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4	-4	-4
				-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4	-4
					-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4	-4
						-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4	-4
							-4	-4	-4	-4	12	12	12	-4	-4	-4	-4	-4
								-4	-4	-4	-4	12	12	12	-4	-4	-4	-4
									-4	-4	-4	-4	12	12	12	-4	-4	-4

Figure 5.58. Initial economic block model.

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
①	-4	-4	-4	-4	-4	0	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
②		-4	-4	-4	-4	0	12	12	8	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
③			-4	-4	-4	-4	8	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4	-4
④				-4	-4	-4	0	12	12	8	-4	-4	-4	-4	-4	-4	-4	-4	-4
⑤					-4	-4	-4	8	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4
⑥						-4	-4	0	12	12	8	-4	-4	-4	-4	-4	-4	-4	-4
⑦							-4	-4	8	12	12	0	-4	-4	-4	-4	-4	-4	-4
⑧								-4	0	12	12	8	-4	-4	-4	-4	-4	-4	-4
⑨									-4	12	12	12	0	-4	-4	-4	-4	-4	-4

Figure 5.59. Final economic block model (Lerchs & Grossmann, 1965).

Row	Current Value	Revised Value
i = 1	12	12
i = 2	12	24 = 12 + 12
i = 3	8	32 = 12 + 12 + 8
i = 4	0	32 = 12 + 12 + 8 + 0
i = 5	-4	28 = 12 + 12 + 8 + 0 - 4
i = 6	-4	24 = 12 + 12 + 8 + 0 - 4 - 4
i = 7	-4	20 = 12 + 12 + 8 + 0 - 4 - 4 - 4
i = 8	-4	16 = 12 + 12 + 8 + 0 - 4 - 4 - 4 - 4

Figure 5.60. Calculation of the cumulative sums for column 6.

where M_{ij} is the profit realized in extracting a single column with block (i, j) at its base and m_{ki} is the net value of block (k, j) . Applying the equation to find the value of the column for $j = 6$, $i = 3$

$$\begin{aligned}
 M_{36} &= \sum_{k=1}^3 m_{k6} = m_{16} + m_{26} + m_{36} \\
 &= 12 + 12 + 8 = 32
 \end{aligned}$$

- ▶ The block positions will be denoted using an i, j numeration system. In keeping with the nomenclature used by Lerchs and Grossmann, i refers to the rows and j to the columns.
- ▶ The first step in this procedure is to calculate cumulative profits for each column of blocks starting from the top and moving downward.

The equation which describes this process is

$$M_{ij} = \sum_{k=1}^i m_{kj}$$

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
①	-4	-4	-4	-4	-4	-4	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
②	-8	-8	-8	-8	-8	8	24	24	8	-8	-8	-8	-8	-8	-8	-8	-8	-8	-8
③	-12		-12	-12	-12	4	32	36	20	-8	-12	-12	-12	-12	-12	-12	-12	-12	-12
④				-16	-16	0	32	48	32	0	-16	-16	-16	-16	-16	-16	-16	-16	-16
⑤					-20	-4	28	56	44	12	-16	-20	-20	-20	-20	-20	-20	-20	-20
⑥						-8	24	56	56	24	-8	-24	-24	-24	-24	-24	-24	-24	-24
⑦							20	52	64	36	4	-24	-28	-28	-28	-28	-28	-28	-28
⑧							16	48	64	48	16	-16	-32	-32	-32	-32	-32	-32	-32
⑨								60	56	28	-4	-32	-32	-32	-32	-32	-32	-32	-32

Figure 5.61. Completed cumulative sums (Lerchs & Grossmann, 1965).

COMPUTER ASSISTED METHODS

- ▶ *The RTZ open-pit generator*
- ▶ *Computer assisted pit design based upon sections*

The RTZ open-pit generator

- ▶ The computer techniques using sections do not solve the problem of end sections or the smoothing of perimeters and consequently have definite limitations. The projection of volumes in the form of cones also has some of the same limitations. The approach described here is the projection of plan areas, specifically the projection of perimeter outlines.
- ▶ The process begins with the development of a block model. Each block as a minimum would be assigned location coordinates and an index character dependent on the rock type or grade type.
- ▶ The second step in the process is the selection of one or more base perimeters from which to generate the pit. These define the final horizontal extent of the pit at, or close to, the elevation of the pit's final base. Each perimeter so drawn becomes the trial base from which an overall pit is generated.
- ▶ The information that is required to carry out these calculations is:
 1. Size and shape of ore body.
 2. Ore and rock types present.
 3. Grades of ore.
 4. Rock slope stability.
 5. Size and shape of base perimeter.
 6. Cutoff grades.
 7. Depth of pit.
 8. Unit working costs.

The trial base perimeter is defined by a series of short chords

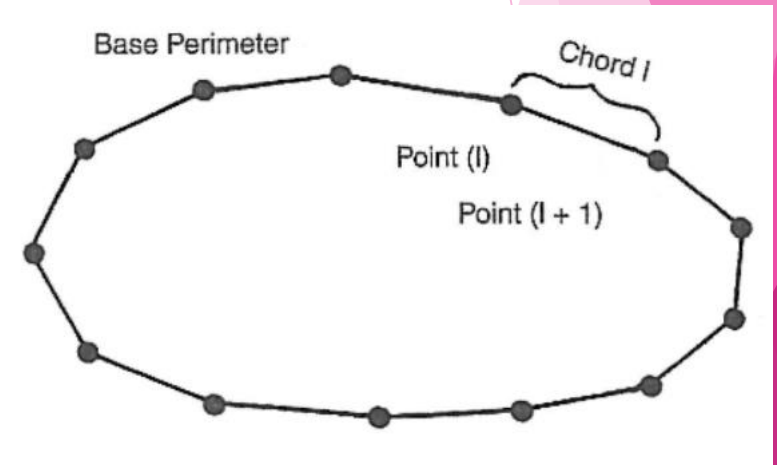


Figure. Trial based perimeter

- ▶ The coordinate pairs H_i , I_i and J_i are projected outwards and upwards to the next upper horizon in a direction normal to the perimeter at that point. The next upper horizon occurs at a specified distance above the current horizon. The actual interval is determined by the requirements of the job at hand and is some fractional part (usually one-half) of the bench height. The slope angle used is that permitted in the particular rock type or at that pit position.

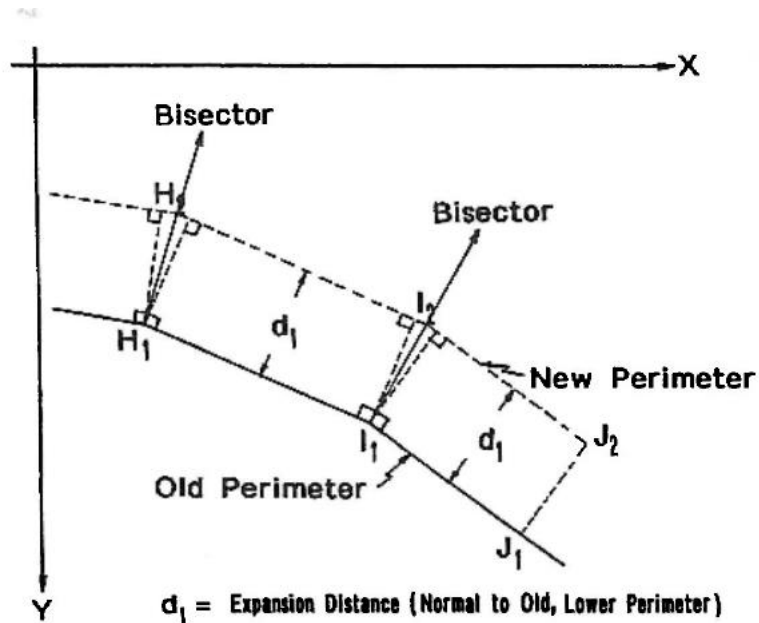


Figure 5.113. Projection of chords and perimeter points along the bisectors

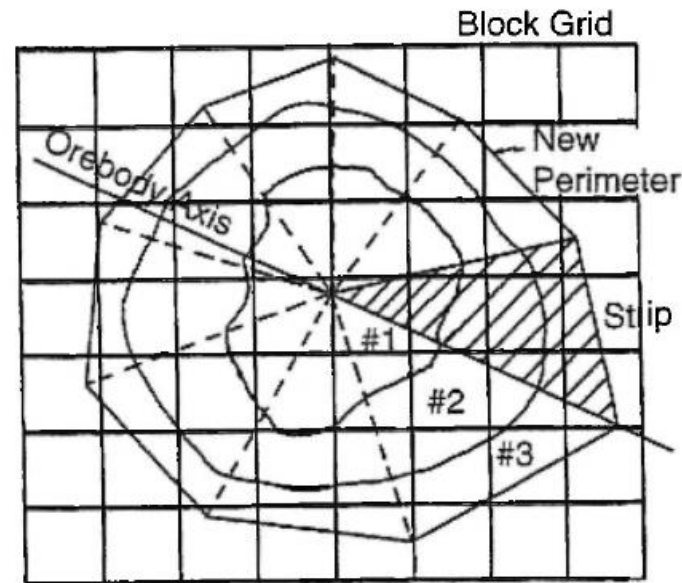


Figure 5.116. Superposition of the bench perimeters on the block model

A working extension to the first pit

An incremental expansion of the pit to test the sensitivity of the first pit to profitability

Computer assisted pit design based upon sections

- ▶ The block size could be made still quite large and yet be small with respect to the size of the deposit and the scale over which the major grade changes take place. The distinction between ore and waste was made on the basis of cut off grade rather than on a sharp boundary.
- ▶ There are a number of applications where block models are less satisfactory. Steeply dipping, relatively narrow vein/strata form deposits being one example. Here the ore is contained within definite, well defined boundaries. Grade variation within the ore zone is such that an average ore grade may be assigned. In such cases, the use of vertical sections still plays an important role. One computer assisted technique for dealing with such situations has been described

The straight line approximations for the slopes are used to simplify the process. Once the 'final' best position is determined then the functional mining parameters such as:

- The ramp(s),
- working bench heights,
- berm widths, and
- bench face angles are added

The design is then reexamined. Often significant changes occur between the simplified and actual pit designs. When the 'final' design on this section has been located, the areas of overburden, ore and waste are obtained by planimetry. Overall stripping ratios, etc. can then be calculated.

Design changes such as varying

- minimum pit width,
- slope angles,
- position/width of the ramp, and
- limiting stripping ratio

would require the entire process to be repeated.

Computer techniques have been developed to assist in the preparation of the sections and in making the necessary calculations. In this way a large number of potential designs can be evaluated quickly and inexpensively

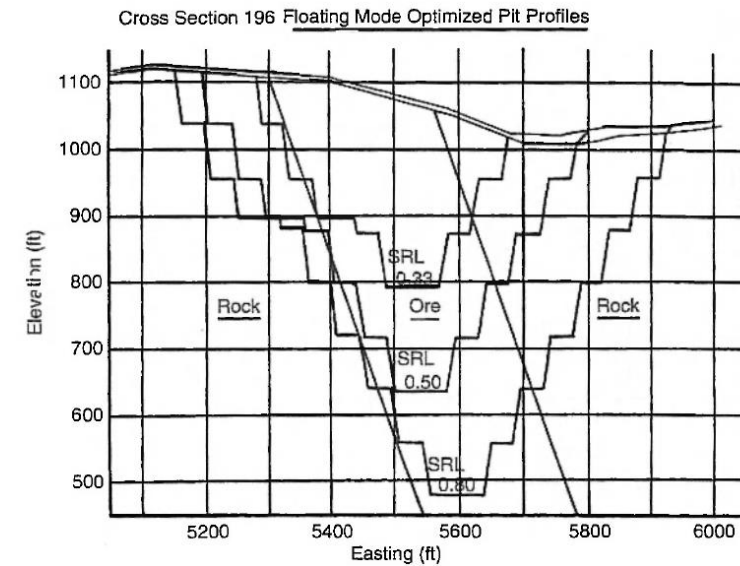


Figure 5.126. Pit profiles for three SRL values (Luke, 1972).

The end points of this segment form the toe positions of the left and right hand slopes. The walls are defined by:

1. Bench height. The bench elevation differential on the ultimate wall.
2. Bench width.
3. Wall angle. The bedrock wall angle from bench to bench, not the overall slope angle.
4. Valid bench elevation. The elevation of any existing berm if the pit is under current development, or the elevation of the first proposed bench. From this specified bench, the elevation of successive benches is determined from the bench height.
5. Overburden angle. The wall angle that can be maintained in overburden.
6. Toe position. The (X, Y) coordinates for the indicated intersection of the pit wall and pit floor.

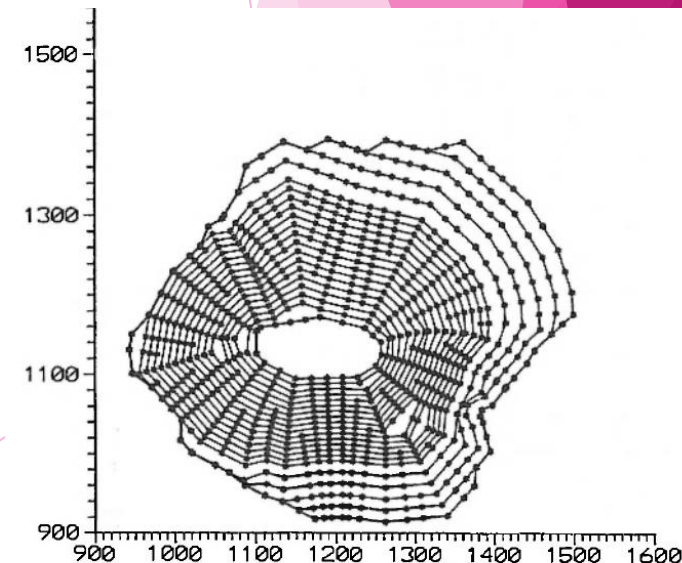


Figure 5.118. A typical pit design (Motta, 1988).

Thanks